

Speed Restrictions

Revision Guide — Rule Book Module SP | EMR 13-05
GERT8000-SP Issue 7 (December 2024) | Sectional Appendix | Weekly Operating Notice

1. Types of Speed Restriction

Speed restrictions are provided to enable train movements to proceed at a safe and efficient speed. There are three categories:

Abbreviation	Definition / Where Published
PSR — Permissible Speed	Published in Table A of the Sectional Appendix. Permanent.
TSR — Temporary Speed Restriction	Details published in the Weekly Operating Notice (WON). Time-limited.
ESR — Emergency Speed Restriction	Imposed without WON publication, or differs from the WON. Communicated directly by the signaller.

The definitions below are taken directly from **Rule Book Module SP Section 1**.

ESR — full rule book definition (non-ERTMS lines)

A speed restriction which:

- has not been published in the Weekly Operating Notice
- has been published, but the times, speed or limits are different from those published
- has been imposed again after being withdrawn early
- has been shown in an amendment to the Weekly Operating Notice

Blanket speed restriction

A speed restriction which **applies to an area** rather than a geographical location.

Differential speeds

If there is a differential permissible speed, or a differential temporary or emergency speed restriction: the **higher speed** applies to passenger, parcels and postal trains (loaded or empty) and light locomotives. The **lower speed** applies to all other trains.

2. Permissible Speeds (PSR)

Permissible speeds control safe train movement across the network. Found in **Table A of the Sectional Appendix**.

2.1 Driver Obligations (Rule Book SP Section 2.1)

Rule Book SP: You must:

- control the speed of your train to **no more than the permissible speeds** on all sections of the line
- make sure the **whole of your train has passed clear** of a section with a lower speed before increasing your speed

Rule Book SP: If the change in permissible speed is marked by an **acceleration indicator**, you can accelerate to the new permissible speed as soon as the **front of the train** is at the position of the sign.

Rule Book SP: Where there are **differential permissible speeds**, control your speed to no more than the speed that applies to **your** train type.

2.2 Train Type Letters — Differential Speed Boards

Letter	Train Type
HST	High Speed Trains
MU	Multiple-unit trains
DMU	Diesel multiple-unit trains
EMU	Electric multiple-unit trains
SP	Sprinter multiple-unit trains
CS	Class 67 locomotives

The **bottom figure** on a differential board is always the higher speed. The **top figure** (lower speed) applies to all other trains.

2.3 Permissible Speed Warning Indicators

On high-speed lines or where there is a significant speed reduction, a warning indicator **may** be provided on approach to the permissible speed indicator. This may sometimes have **associated AWS** to warn drivers of the speed reduction.

2.4 Diverging Junctions

At diverging junctions, the PSR for the diverging route is indicated by a board with a **direction arrow**. On higher-speed routes an additional PSR may be used prior to the junction. **Your route knowledge is key.**

2.5 Enhanced Permissible Speed (EPS)

Used on routes operated by **tilting trains** (Class 221 and Class 390). Where differential signs are provided, the bottom figure shows the higher speed and applies to Class 390 in tilting mode; the top figure applies to Class 221 in tilting mode.

2.6 Locomotives Running Light or Hauling Trains (Rule Book SP Section 2.2)

These restrictions apply **only when the permissible speed is more than 60 mph (95 km/h)**. These are speed limits for specific loco formations:

Table 1 — Loco light or short loco-hauled formations:

Any number of locos light; or 1–2 locos + 1–3 coaching stock; or 3+ locos + any coaching stock	75 mph (120 km/h)	60 mph (95 km/h)

Table 2 — Locomotive-hauled trains with coaching/postal/parcels vehicles (where permissible > 75 mph):

Train formation	≥ 100 mph (160 km/h)	90–95 mph (145–155 km/h)	80–85 mph (130–135 km/h)
1 loco + 4–6 vehicles, or 2 locos + 4–10 vehicles	90 mph (145 km/h)	80 mph (130 km/h)	75 mph (120 km/h)

You **do not need to apply** these restrictions to locomotive classes or train formations listed in your TOC instructions that have sufficient brake power.

DRIVER REMINDER — PSR
✓ Permissible speeds are in Table A of the Sectional Appendix
✓ Whole train must clear a lower-speed section before increasing speed
✓ Acceleration indicator: you can increase speed when the FRONT of the train reaches the sign
✓ Bottom figure on differential board = higher speed (passenger, parcels, postal, light loco)
✓ Check loco running light / hauled train speed restrictions if applicable

3. Temporary Speed Restrictions (TSR)

TSRs are imposed for infrastructure maintenance or other safety reasons. Details are **published in the WON**. **Check the WON and notices when booking on.**

3.1 Driver Obligations (Rule Book SP Section 3.1)

Rule Book SP: When driving over a TSR, you must:

- control the speed of your train to **no more than the speed shown on the warning board**
- make sure the **whole of your train has passed clear** of the restricted section before increasing speed

Rule Book SP: Where there are differential speeds, control your speed to no more than the speed that applies to **your** train.

3.2 Normal TSR Equipment Sequence (Rule Book SP Section 3.2)

Equipment	Position / Rule
Portable AWS magnet	Normally 180 metres (approx. 200 yards) on approach to the warning board
Warning board	Placed on the approach to the speed indicator at the service braking distance
Speed indicator	At the start of the TSR — speed must be at or below this figure before the front of your train reaches it
Termination indicator	At the end of the TSR — do not increase speed until the rear of your train has passed this
(Alternative) Acceleration indicator	May replace termination indicator — can accelerate when the front of the train reaches it

CRITICAL: Any permissible speed boards showing a higher speed within the TSR must be DISREGARDED.

3.3 TSR on Single and Bi-directional Lines (Section 3.5)

TSR equipment must be provided **in both directions** on a single or bi-directional line.

After passing through a TSR, you will receive an AWS warning from the other-direction warning board. A **cancelling indicator** is normally placed **180 metres (approx. 200 yards)** beyond that AWS magnet (minimum 45 metres / 50 yards).

The cancelling indicator confirms you can **disregard** the AWS warning just received.

3.4 Consecutive TSRs (Section 3.6)

a) Lower speed TSR immediately beyond a higher speed TSR

- No termination indicator at the end of the first (higher) TSR
- A speed indicator showing the lower speed is provided instead
- Only one termination indicator — at the end of the second TSR

b) Higher speed TSR immediately beyond a lower speed TSR

- No warning board for the second (higher speed) TSR
- No termination indicator at the end of the first (lower speed) TSR — replaced by a speed indicator showing the higher speed
- Only one termination indicator — at the end of the second TSR

c) Insufficient distance to position boards normally

- Second warning board placed at least **45 metres (approx. 50 yards)** beyond the first
- Second portable AWS magnet placed immediately beyond the first warning board

3.5 TSR Within a TSR (Section 3.7)

a) Lower speed restriction **INSIDE** a higher speed restriction

- Equipment provided normally, except: no termination indicator at the end of the lower speed section
- Speed indicator showing the outer (higher) speed placed at that point instead
- One termination indicator at the end of the outer TSR

b) Higher speed restriction **INSIDE** a lower speed restriction

- No warning board for the inner (higher speed) TSR
- No termination indicator at the end of the lower speed section — speed indicator showing the higher speed instead
- One termination indicator at the end of the outer TSR

3.6 TSR at a Diverging Junction (Section 3.8)

Where a TSR applies to the diverging route only, the **warning board and speed indicator both carry a direction indicator** to show which route the TSR applies to.

3.7 TSR Beyond a Station or Siding Connection (Section 3.9)

If the speed indicator is more than **300 metres (approx. 330 yards)** beyond a passenger station, siding connection or dead-end platform, a **repeating warning board** is placed as a reminder. It is positioned:

- next to the platform starting signal (if there is one), or
- next to the siding exit signal, or
- immediately ahead of the station or siding connection

A portable AWS magnet is **not** provided on the approach to the repeating warning board.

3.8 TSR at a Location Where Trains Can Reverse or Change Drivers (Section 3.10)

An **additional speed indicator** is placed within the TSR as a reminder. It is positioned:

- next to the starting signal, or
- immediately ahead of the station

3.9 Spate Indicators — When a TSR is Not Introduced or Removed Early (Sections 3.13–3.14)

A **SPATE indicator** is displayed when:

- A published TSR is **not being imposed** (but 24 hours' notice was not possible before the start time), or
- A TSR has been **removed earlier than published** in the WON

When SPATE is shown, the **normal permissible speed applies**. The warning boards and indicators remain in position for the planned duration.

When a TSR is **eased** (higher speed than published), the warning boards and speed indicator are **changed to show the higher speed**.

3.10 When a TSR Becomes an ESR (Rule Book SP — Definitions, page 4)

A restriction is a TSR only as long as its details are published accurately in the WON. It becomes an **Emergency Speed Restriction** if:

- it has **not been published** in the Weekly Operating Notice
- it has been published, but the **times, speed or limits differ** from those published
- it has been **imposed again after being withdrawn early**
- it has been shown in an **amendment to the Weekly Operating Notice**

At that point the restriction no longer meets the TSR definition — it is an ESR, and ESR procedures apply. The signaller will stop you and tell you the location and speed. When equipment is later put in place, an **emergency indicator** will be provided on approach to the warning board.

DRIVER REMINDER — TSR
✓ Check the WON when booking on — know all active TSRs on your route
✓ AWS magnet = normally 180m (200 yds) before the warning board
✓ Reduce speed BEFORE the front of your train reaches the speed indicator
✓ Ignore any PSR boards showing a higher speed within the TSR
✓ Rear of train must clear termination indicator before increasing speed
✓ Speed increases to lower of: line permissible speed OR your traction type maximum
✓ On bi-directional lines: cancelling indicator (180m past the AWS) = disregard that AWS warning
✓ Repeating warning board after a station = no AWS on approach to it
✓ SPATE indicator = TSR not in force, normal permissible speed applies

4. Emergency Speed Restrictions (ESR)

ESRs are imposed **without WON publication**, or differ from published information. They require immediate action.

4.1 Being Stopped Before Equipment is in Place (Section 4.2)

If you must pass over an ESR **before trackside equipment is in place**, the signaller will stop you and tell you:

- the location where the ESR **begins and ends**
- the **speed limit** imposed

Rule Book SP: You must:

- control the speed of your train to travel over the affected line at **no more than the speed the signaller tells you**
- make sure the **whole of your train has passed clear** before increasing speed

4.2 ESR Equipment — Normal Arrangements (Section 4.3)

When an ESR is to last more than a short time, full equipment is provided as soon as possible. This includes all normal TSR equipment **plus an emergency indicator**.

Equipment	Position
Portable AWS magnet (for emergency indicator)	Normally 180m (200 yds) on approach to the emergency indicator
Emergency indicator	At least 180m (200 yds) and not more than 400m (440 yds) on approach to the warning board
Portable AWS magnet (for warning board)	Placed at or beyond the emergency indicator
Warning board, speed indicator, termination indicator	As per normal TSR arrangement (Section 3.2)

The emergency indicator has **two flashing white lights** against a **yellow and black chevron** background.

The emergency indicator will **stay in position** until (Section 4.5):

- the details of the speed restriction appear in the Weekly Operating Notice, or
- the speed restriction is withdrawn

4.3 Blanket Speed Restrictions (Section 6)

When a blanket speed restriction is imposed over an **area**:

- Emergency indicators and other track equipment are **not** provided
- The signaller will inform you of the location and speed of the restriction
- Information may also be in the **Late Notice case**

Rule Book SP: You must control the speed of your train to **no more than the restriction throughout that defined area**.

DRIVER REMINDER — ESR
✓ Signaller stops you and gives you start/end location and speed before equipment is in place
✓ Emergency indicator = 2 flashing white lights on yellow/black chevron — positioned 180–400m before warning board
✓ Emergency indicator stays until ESR is in WON OR restriction is withdrawn
✓ Blanket ESR: no trackside equipment — signaller and Late Notice case are your sources
✓ EROS = Emergency Restriction of Speed

5. Defective or Missing ESR/TSR Equipment

5.1 Warning Boards or Speed Indicators Missing or Incorrect (Section 5.1)

Rule Book SP: You **must tell the signaller immediately** if a warning board, repeating warning board or speed indicator is:

- **missing**
- in a **different place** from that published in the WON
- **more restrictive** than shown in the WON

Rule Book SP: If necessary, you must **stop your train specially**.

You do not have to report this if you have already been told about it.

5.2 Equipment Difficult to See (Section 5.2)

Rule Book SP: If you see a warning board, repeating warning board or speed indicator that **is, or is becoming, difficult to see**, you must tell the signaller **at the first opportunity**.

5.3 Defective or Missing Emergency Indicator (Section 5.3)

Rule Book SP: If you see anything wrong with the emergency indicator, you must **tell the signaller immediately** — stopping your train if necessary.

CRITICAL: If boards are missing, wrong, more restrictive than the WON, or the emergency indicator is defective — tell the signaller immediately. Stop your train if necessary.

6. Revision Questions & Answers

All answers sourced from Rule Book GERT8000-SP Issue 7 and EMR PPT 13-05.

Q: What are the three types of speed restriction?

A: PSR (Permissible Speed Restriction) — Sectional Appendix; TSR (Temporary Speed Restriction) — published in the WON; ESR (Emergency Speed Restriction) — not in WON or differs from it.

Q: On a differential speed board, which figure is the higher speed and who does it apply to?

A: The BOTTOM figure is always the higher speed. It applies to: passenger trains, parcels and postal trains (loaded or empty), and light locomotives.

Q: When can you increase speed after passing a permissible speed board showing an increase?

A: Not until the FULL LENGTH of your train has passed the permissible speed board. Exception: if an acceleration indicator is provided, you can accelerate once the FRONT of your train is at the sign.

Q: How far before a TSR warning board is the portable AWS magnet normally placed?

A: 180 metres (approximately 200 yards) on approach to the warning board.

Q: Before what point must you reduce to TSR speed?

A: Before the FRONT of your train reaches the speed indicator board at the start of the TSR.

Q: When can you increase speed after a TSR?

A: Not until the REAR of your train has completely passed the termination indicator. Then increase to the lower of: the permissible line speed, or your traction type maximum.

Q: You have passed through a TSR but the termination board is missing. What do you do?

A: Tell the signaller immediately. If necessary, stop your train.

Q: What is a spate indicator and when is it used?

A: It is displayed when a published TSR has not been imposed (because insufficient notice was given) or has been removed early. When SPATE is shown, the normal permissible speed applies.

Q: What does a directional indicator on a TSR warning board mean?

A: The restriction applies to another line — to the left or right of the line you are on.

Q: On a bi-directional line, after passing through a TSR you receive an AWS warning. What do you do?

A: Wait for the cancelling indicator, normally placed about 180m (200 yds) beyond that AWS magnet. When you see it, you can disregard the AWS warning you received.

Q: In a consecutive TSR (lower speed immediately beyond higher speed), where is the termination indicator?

A: There is only ONE termination indicator, at the end of the SECOND (lower speed) TSR. No termination indicator is placed between the two restrictions — a speed indicator showing the lower speed replaces it.

Q: A TSR is within another TSR (lower inside higher). Where is the termination indicator?

A: One termination indicator at the end of the OUTER (higher speed) TSR. The inner TSR ends with a speed indicator showing the outer speed instead of a termination indicator.

Q: When is a repeating warning board provided for a TSR?

A: When the speed indicator is more than 300 metres (approx. 330 yards) beyond a passenger station, siding connection, or dead-end platform. No portable AWS magnet is provided on approach to the repeating warning board.

Q: What is placed inside a TSR at a location where trains can reverse or change drivers?

A: An additional speed indicator is placed within the TSR as a reminder, next to the starting signal or immediately ahead of the station.

Q: What is the appearance of an emergency indicator?

A: Two flashing white lights against a yellow and black chevron background. It is preceded by an AWS warning.

Q: Where is the emergency indicator positioned relative to the warning board?

A: At least 180 metres (200 yards) and not more than 400 metres (440 yards) on approach to the warning board.

Q: How will you be informed of an ESR before equipment is in place?

A: The signaller will stop you and tell you where the ESR begins and ends, and the speed to observe.

Q: When does the emergency indicator come down?

A: When the ESR details appear in the Weekly Operating Notice, or the speed restriction is withdrawn.

Q: Equipment at an ESR is showing a more restrictive speed than published in the WON. What do you do?

A: Tell the signaller immediately — stopping your train if necessary.

Q: What is a blanket speed restriction and how are you informed?

A: A restriction applied to an area rather than a specific location. No trackside equipment is provided. The signaller tells you the location and speed, and information may be in the Late Notice case.

Q: What does EROS stand for?

A: Emergency Restriction of Speed.

Q: What are the speed limits for any number of locos running light where the permissible speed is 90 mph or above?

A: 75 mph (120 km/h). Where the permissible is 85 mph or less, the limit is 60 mph (95 km/h).

Q: Define an ESR according to Rule Book Module SP.

A: A speed restriction which: has not been published in the WON; or has been published but the times, speed or limits are different from those published; or has been imposed again after being withdrawn early; or has been shown in an amendment to the WON.